E9 205 Machine Learning for Signal Processing

Dimensionality Reduction - I

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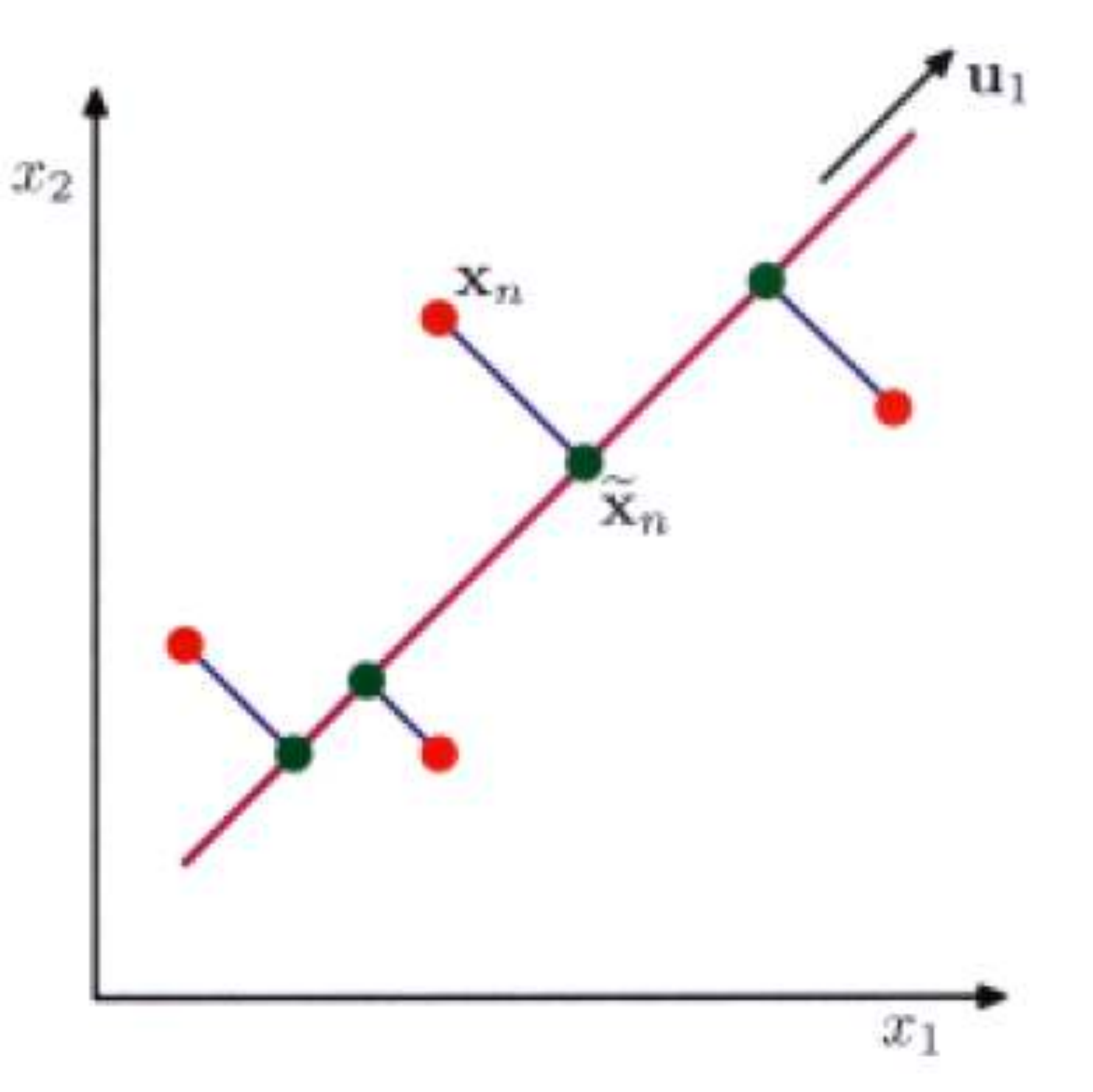
Principal Component Analysis

- Reducing the data $\mathbf{x}_n$ of dimension D to lower dimension $M < D$
- Projecting the data into subspace which preserves maximum data variance
  - Maximize variance in projected space
- Equivalent formulated as minimizing the error between the original and projected data points.

Minimum Error Formulation - PCA



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Principal Component Analysis

- First M eigenvectors of data covariance matrix

$$S = \frac{1}{N}\sum_{n=1}^{N}(\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T$$

- Residual error from PCA

$$J = \sum_{i=M+1}^{D} \lambda_i$$

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PCA

Handwritten digits used for PCA training…

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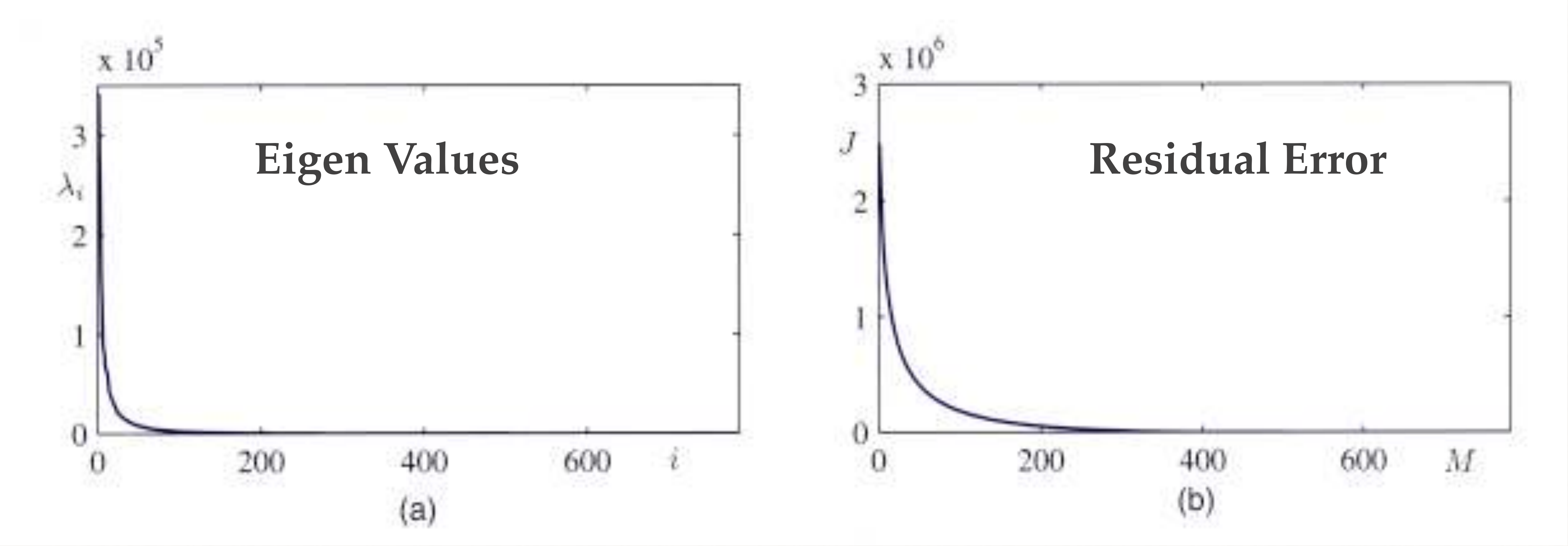
PCA



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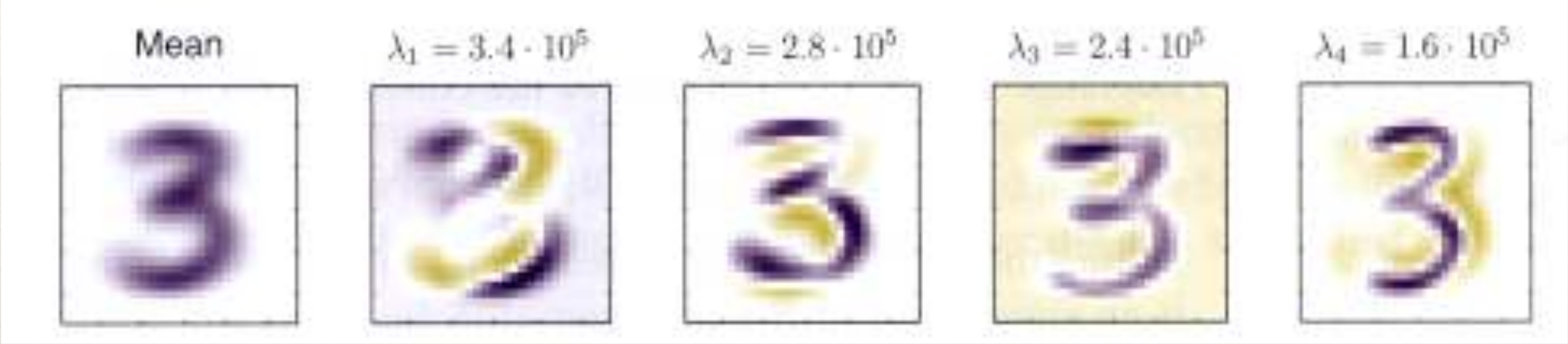
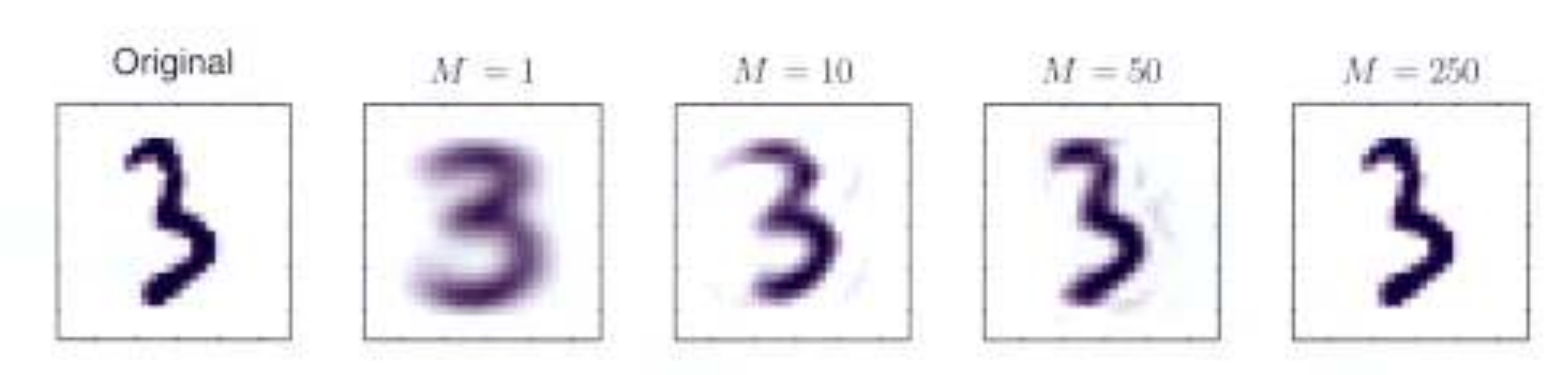
PCA - Reconstruction

Eigenvectors



PCA - Reconstruction



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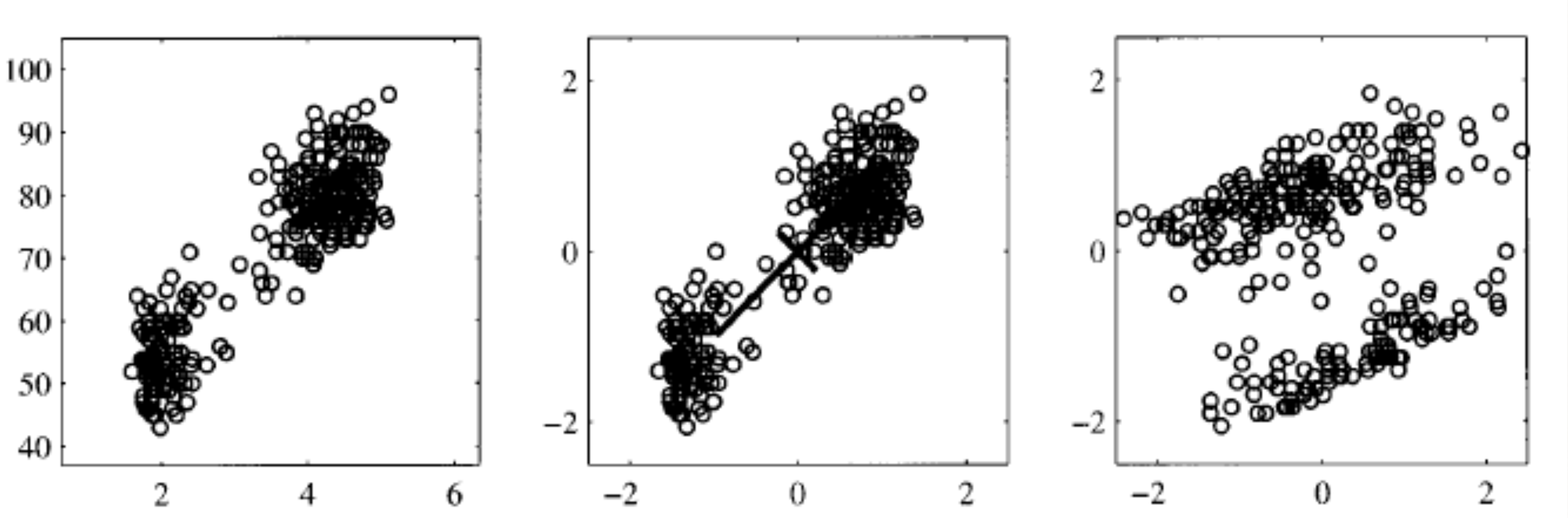
Whitening the Data

Original Data | **Whitened data** | **Whitened data**



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